

CINDER v1.1.2 energy audit — interpretation memo

Interpretation memo: CINDER v1.1.2 energy-consistency study

Meaning of the accounting:

External work = cumulative net work crossing the modeled shaft boundaries. It is energy injected into, or extracted from, the audited system by the boundary torques. A resisting load contributes negative boundary work.

Stored energy = energy currently resident in modeled energy stores/states, i.e. kinetic-energy terms plus retained potential/spring terms.

Slip dissipation and impact/capture dissipation are irreversible channels.

Nominal final balance:

External work	=	52.168893 kJ
Stored-energy change	=	50.580274 kJ
Slip dissipation	=	1.588340 kJ
Impact/capture dissipation	=	0.253859 J
Residual	=	0.025444 J
Residual / external work	=	4.877154e-07

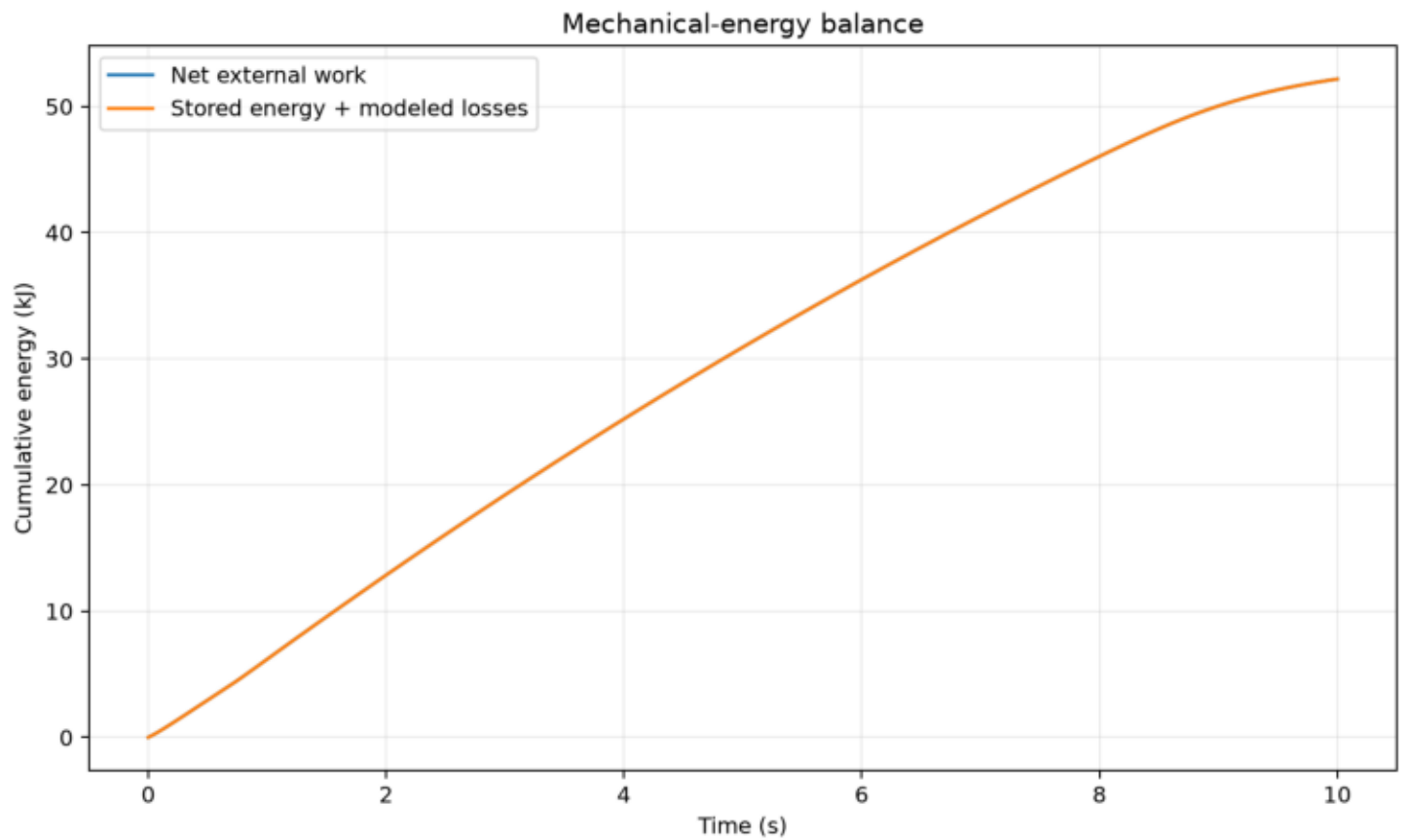
Key interpretation:

- The earlier 15 J structural discrepancy is gone in v1.1.2.
- Quadrature Richardson estimate = 0.024510 J
- Worst event-energy defect = 2.274e-13 J
- Tight final-state delta = 1.298e-07
- Tight residual = -0.022843 J
- Remaining residuals are at numerical-error scale.

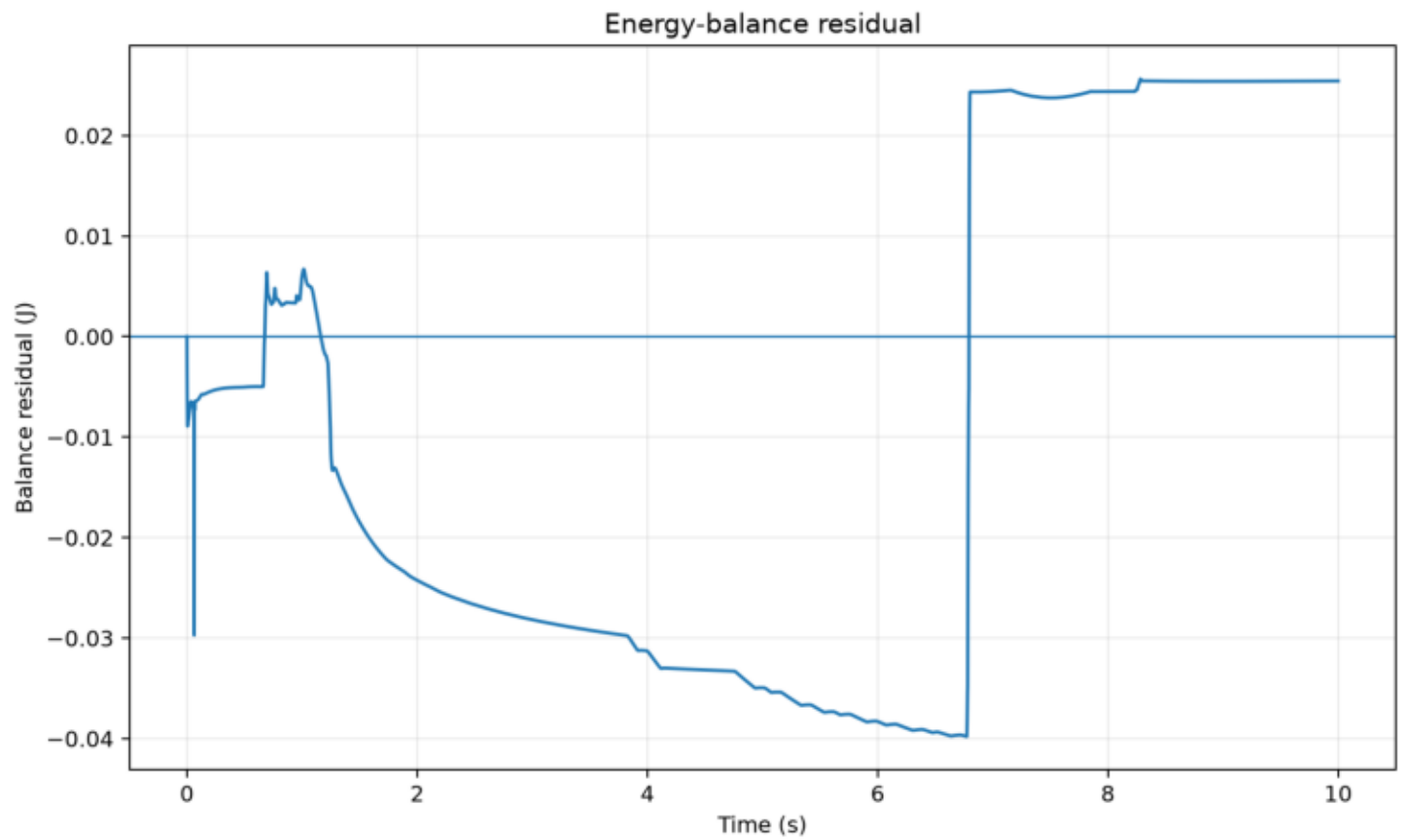
Status:

This study looks strong enough to archive now and return to later when the paper results section is written.

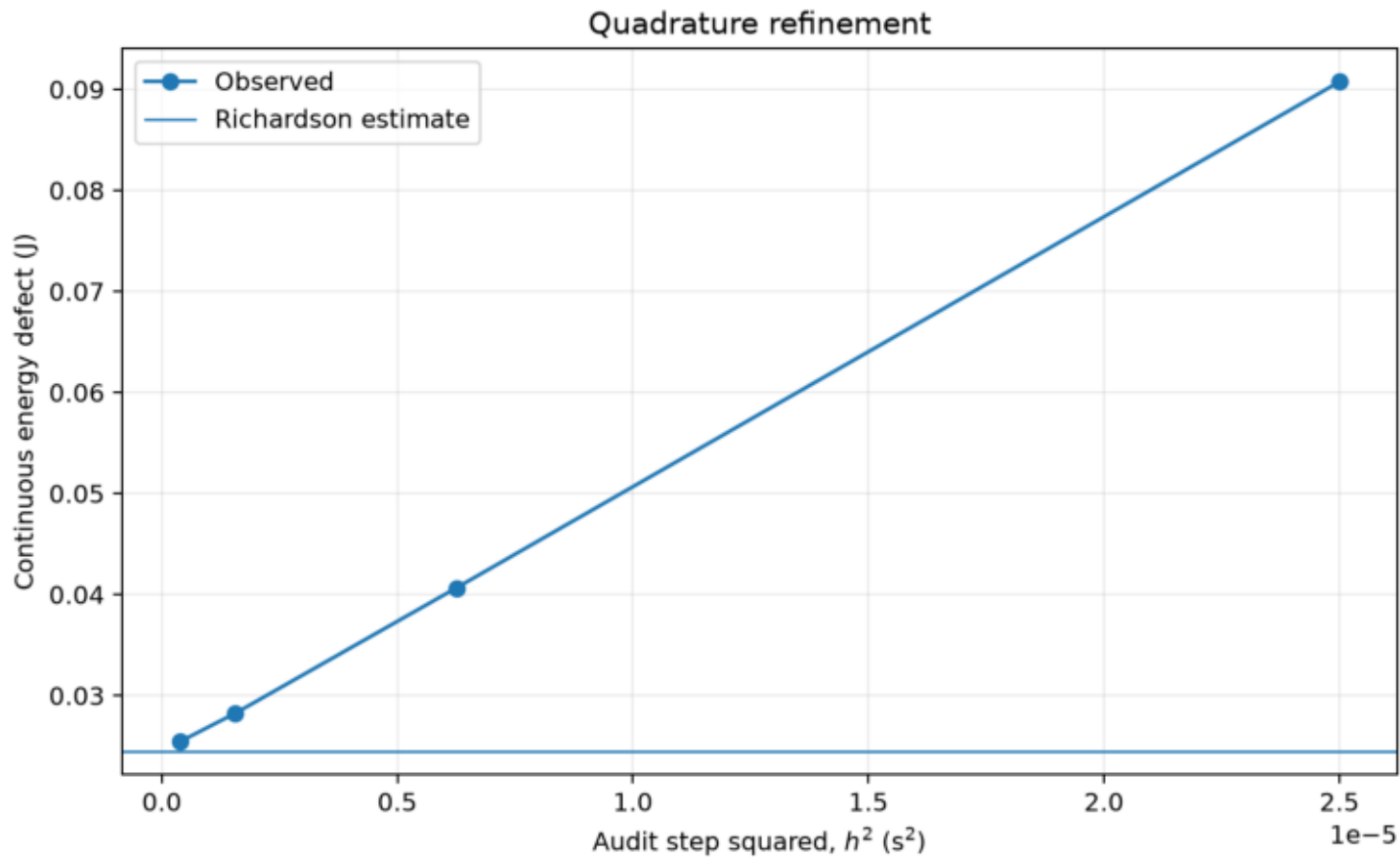
Cumulative energy balance



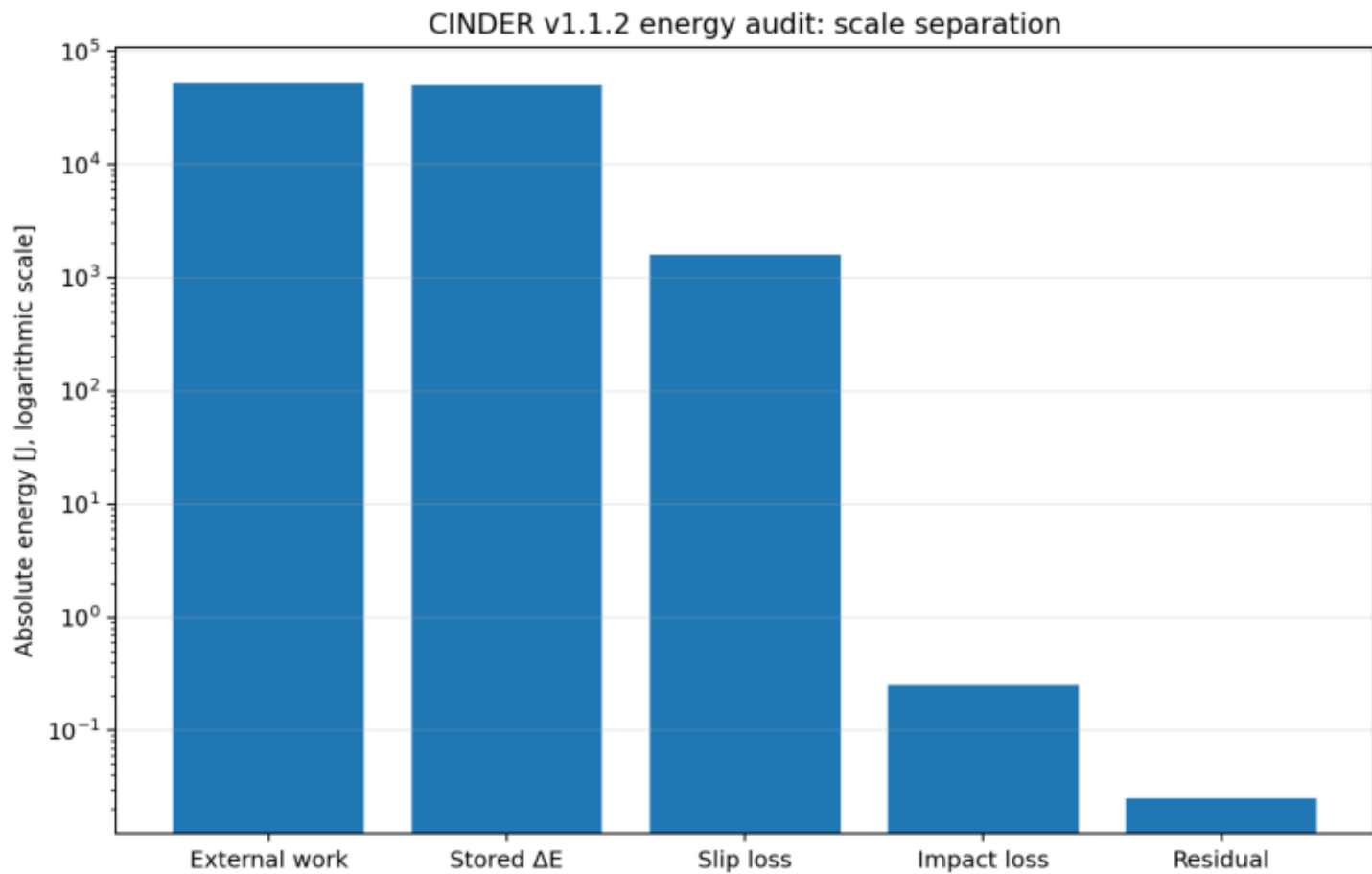
Residual energy balance



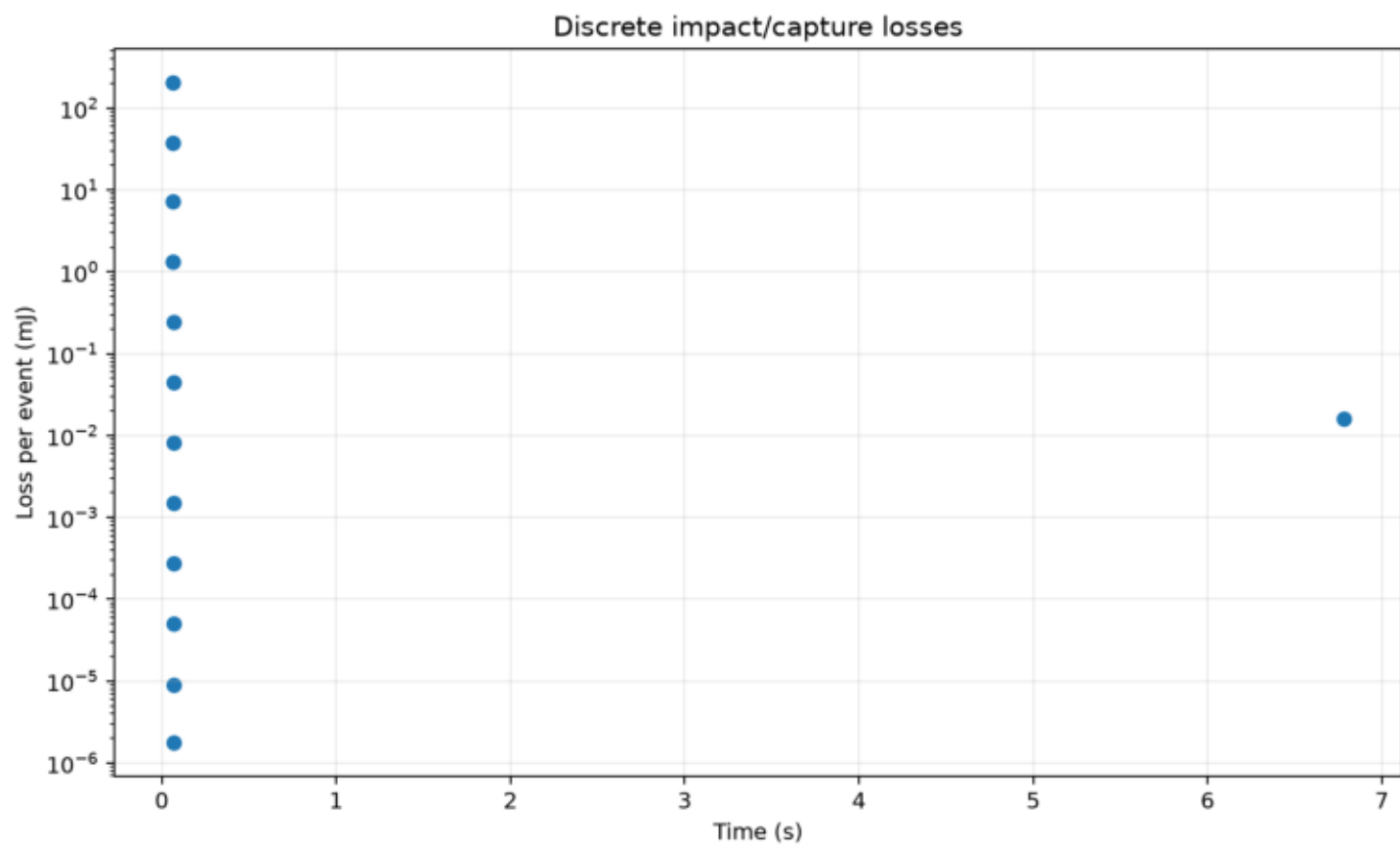
Quadrature convergence



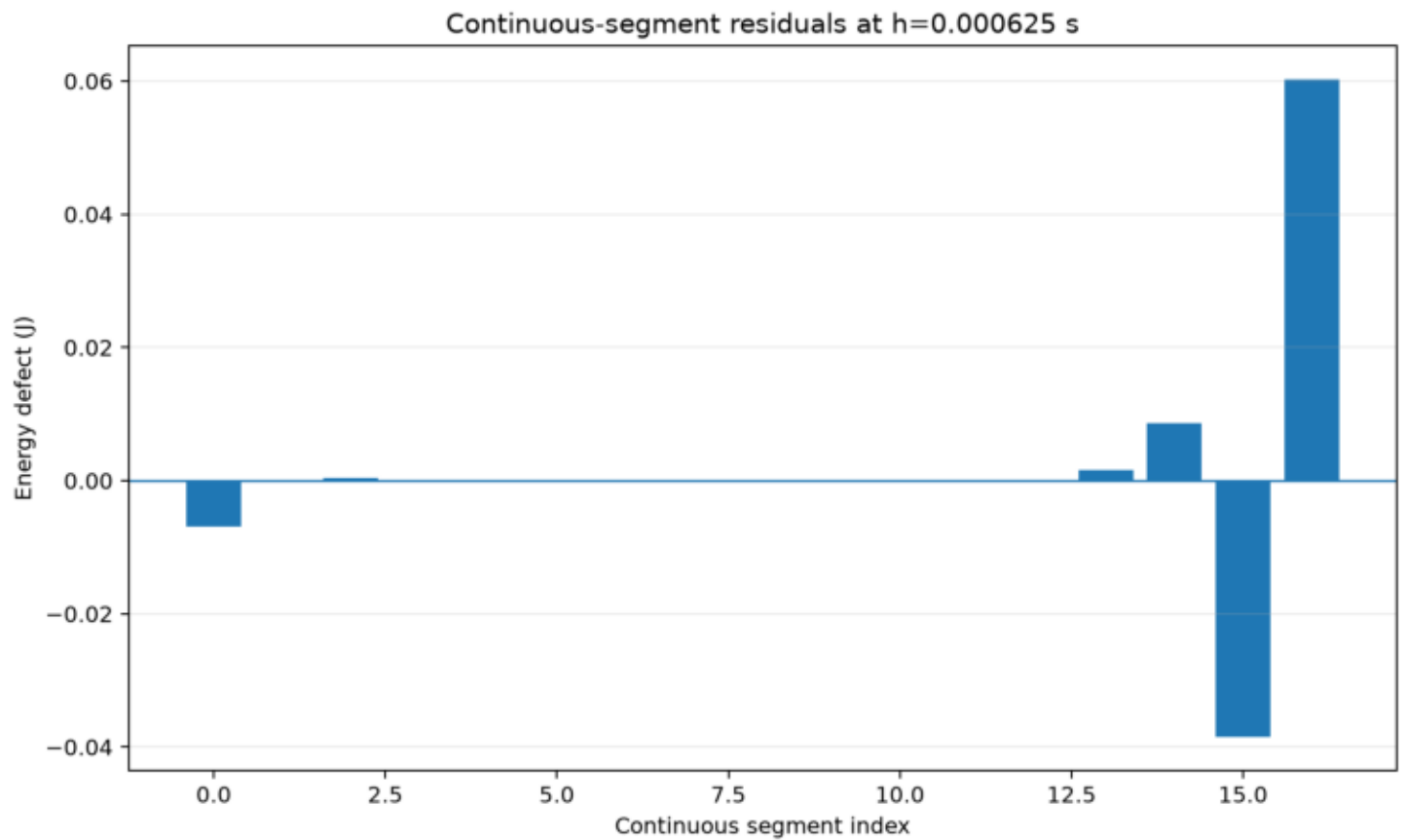
Energy scale separation



Discrete event/capture losses



Continuous segment residuals (supporting audit view)



Energy channels (supporting view)

